Link-loss response as a UAS fingerprint - project proposal

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1 Research problem

To what extent can UAS be indentified by fusing physical layer and network layer behavioral responses during C2 link disruption?

The experiment will test hypotheses H1-H4 regarding stimulus response and multimodal fusion.

Table 1: Evaluation Hypotheses

Hypothesis	Description
H1	Identification from the window after a link loss is more accurate than from a steady-flight window of the same length
H2	The gain from fusing RF and network features is largest in the response window
H3	The link-loss response is distinguishable from the response to benign congestion and weak radio
H4	The fingerprint survives when failsafe timeouts are set equal across profiles

2 Theory

The way a UAS reacts when having its command and control link disrupted is caused by a combination of hardware and software factors. Such factors include how the UAS is configured, as well as which ground station software, on-board software and radio hardware is used.

3 Method

The project relies on a testbed where a virtual UAS system is constructed by combining ground station software, onboard software and a radio link simulation stack. The modules proposed for the testbed are listed below.

Table 2: Simulation modules and purposes

Simulation software	Role
Gazebo	Physics simulation
OMNeT++ / INET	Network simulation
Simu5G	Radio environment simulation

The simulation will be conducted with combinations of 4 factors under test: profile, disruption, radio background and seed. In total They will generate 216 simulation runs. In addition to the main simulation runs

a pilot phase of 48 runs and 18 extra runs are planned. Different ground station and flight software stacks will be evaluated as different profiles.

The metrics being measured at the network layer are traffic burstiness, inter-arrival time and retransmission cadence. At the physical layer metrics are airtime, transmit power and signal-to-noise ratio. These will be compared across a 4-segment run structure (steady flight, disruption, response, recovery).

To verify the simulated radio response a short experiment is planned at ECH. The experiment should connect a UAS radio to a measurement setup that will record I/Q samples and measure the received power from a UAS when the C2 link is disrupted. For this test a Software Defined Radio (SDR) will be used to record the I/Q samples and measure the received power. The radio should be connected through a cable to not emit any radio frequency signals. The cabled connection must use an attenuator to protect the receiving radio hardware. The I/Q samples must be time synchronized with network traffic captures to correlate physical layer changes with network layer link disruption.

4 Time plan

The thesis is conducted over a full semester at 50% speed. The spring semester 2027 runs from week 3 to week 22 inclusive, 20 weeks in total. Below is a proposed time plan from the start of the thesis.

Table 3: Initial project time plan

Weeks	**Planned activity**
1-2	Literature review & problem formulation
3-5	Simulation setup & environment verification
6-9	Simulation run campaign
10-11	Mid term review & RF mesurement setup
12-13	Simulation run campaign
14-15	Data validation & model training
16-18	Report drafting
19-20	Review and final editing